

Example

We've created a preliminary circuit for our product. To ensure its success, we carefully studied the potential risks associated with the design and made necessary adjustments to our original plan. Now, let's focus on exploring the control and user interface aspects of the product.

Determine the CR

Let's take a look at our QFD analysis (February 2023 EFY). This will give us the initial list of requirements. Along with that list, we also need to include the requirements we gathered from the risk analysis (FMEA) and the DFM analysis.

Once we have compiled all the CRs (customer requirements), we can analyse them to map the FRs (functional requirements) and the DPs (design parameters) in order to create the design matrix (Table 2). In this matrix, we might notice that some entries have shared mapping. To organise it better, we will rearrange the design matrix to put it in a diagonal and upper triangular form. After the rearrangement, the matrix will look like Table 3.

The design matrix has two parts. The lime-colour-shaded

Design Parameter	Code	Decision
Controller	DP2	ESP 12F
Relay specification	DP1	A31ACQ24VDC1
Brake relay driver	DP14	BC148
Dummy load relay driver	DP15	BC148
Ambient weather sensor	DP5	DHT11
Dead CPU alarm delay	DP16	555 and buzzer
Env. norm	DP23	No lead soldering
Labour practice	DP24	Fair employment practice
Storage specification	DP7	32GB SD card
Storage module	DP6	SPI SD card module
NTC sensor	DP9	10D9 thermistor
Overheat alarm	DP17	BC148 and LED
Enclosure NEMA standard	DP3	NEMA 3R
PCB design	DP8	Glass epoxy PCB
Enclosure material	DP10	MS for terminal area, acrylic for Wi-Fi area
Enclosure colour	DP11	Black
Console design	DP12	Minimalistic on the box, HTML form for remote access
Hall effect current sensor	DP4	WCS-1600 Hall Effect current sensor
Wi-Fi module	DP13	ESP 12F
Connection status indicator	DP18	LED in controller, on soft interface
Org for after sales	DP19	Web-based CRM
Org for customer support	DP20	Answering machine for tech support
Enclosure size	DP21	250 x 200 x 80 mm ³
User manual	DP22	Online manual
Power supply design	DP25	DC-DC converters

part is a diagonal matrix, and we can choose its design parameters without worrying about other parts. The pink-shaded part forms a sort of a triangular matrix, where the parameters affect various functional requirements, but their relationships follow a hierarchy. To organise our decision-making process, we can

use a design tree (Fig. 4). This tree helps us set the order in which we make our design decisions.

When creating the decision tree, we follow the order of the decision matrix while considering which decisions are more important. The design matrix serves as a helpful guide. The decision tree illustrates the technical hierarchy of the decisions. Here, we discovered that we missed out on deciding on the box material. We need material that can withstand high temperature and occasional flashes in the terminal connection part. Mild steel is a good choice for that application. But we need a non-metallic enclosure for the Wi-Fi module. In this section, we shall use acrylic. We also decided on the mode of after-sales support and the user manual.

In Table 4, you can find the ultimate design parameters we chose for the charge controller.

QFD analysis and the axiomatic design approach share similar scopes. In QFD analysis, design parameters are obtained through benchmarking, while the axiomatic process examines intrinsic dependencies to derive design parameters. When we apply the axiomatic process after QFD, it not only validates the initial design but also helps uncover any missed details from the QFD analysis.

The axiomatic process is a great way to organise the logical steps in software development. Many of our design goals will be achieved through software. The axiomatic decision tree and decision matrix help establish the sequence of actions to be taken in the program. **EFY**

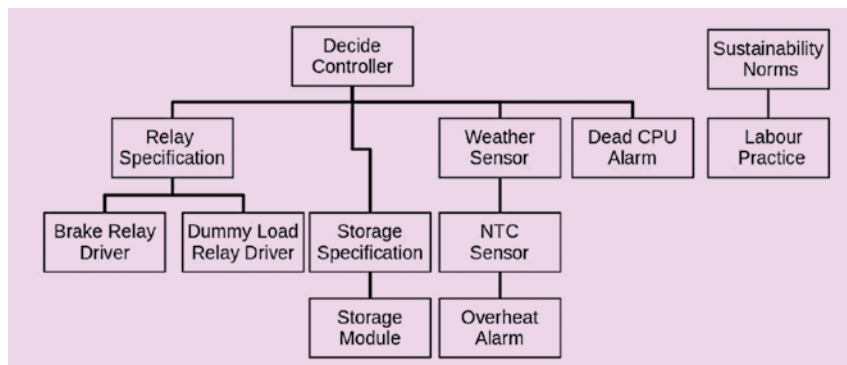


Fig. 4: Decision tree